

Bhumika Mittal

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RESEARCH STATEMENT

My research focuses on building safe and guaranteed systems. In particular, I work on problems in **cryptography**, **programming languages**, and **formal verification**, with a current focus on program analysis for finite field programs. I am also interested in the history of mathematics and science pedagogy.

EDUCATION

Georgia Institute of Technology

Ph.D. in Computer Science

Atlanta, US

Aug 2026 – Present

Ashoka University

Postgraduate Diploma in Advanced Studies and Research; CGPA: 3.93/4.00

Sonipat, India

2024 – 2025

Thesis: Ring Trapdoor Functions: A Lattice-Based Framework for Secure Ring Signatures

Bachelor of Science (Honours), Computer Science; CGPA: 3.90/4.00

2021 – 2024

Minors in Mathematics, and Entrepreneurial Leadership & Strategy

RESEARCH AND WORK EXPERIENCE

Amuse Labs

Software Engineer

Bengaluru, India

July 2025 – June 2026

- Designing a verification endpoint that validates puzzle invariants, ensures correctness, and supports quality checks
- Improving the automated testing tool Gandalf to expand code coverage and validate system correctness
- Developed the Word Flower playlog and hint features for PuzzleMe, a digital platform for smart games, prioritizing type safety, documentation, and maintainability

Max Planck Institute for Software Systems

Visiting Research Fellow

Saarbrücken, Germany

May 2024 – Aug 2024

- Benchmarked various data structures for diverse graph workloads like analytics, traversals, and pattern matching
- Designed PACT: a metric to measure how far a system deviates from an ideal phase-concurrent execution
- Implemented a dynamic graph store with selective snapshot versioning that decouples readers from writers, achieving 72% reduction in read latency and 31% increase in throughput compared to lock-based approaches

Ashoka University

Research Intern

New Delhi, India

Nov 2023 – May 2024

- Designed and analyzed lattice-based schemes for quantum-safe migration of communications systems
- Implemented a constant-time cryptographic module now in active use
- Authored a comprehensive technical report evaluating the security properties, performance, and implementation feasibility of proposed post-quantum cryptographic primitives

IIT Gandhinagar

Research Assistant

Gandhinagar, India

May 2023 – Sept 2023

- Built a tool to model data flow in computation graphs for memory hierarchy evaluation and hardware optimization
- Designed an accelerator architecture, achieving 9x energy and 58% area reduction compared to existing systems
- Analyzed transformer architectures to identify hardware-level optimizations, enabling efficient deployment of language models on edge devices

PUBLICATIONS

AxLaM: Energy-Efficient Accelerator Design for Language Models for Edge Computing

Philosophical Transactions A

Tom Glint, Bhumika Mittal, Santripa Sharma, Abdul Ronak, Abhinav Goud, Neerja Kasture, Zaqi Momin, Aravind Krishna, Joycee Mekie

[Link to paper](#)

On the Existence of Balanced Generalized de Bruijn Sequences

Discrete Mathematics Journal

Matthew Baker, Bhumika Mittal, Haran Mouli, Eric Tang

[Link to paper](#)

HONORS AND AWARDS

2025 **Academic Excellence Award**, Department of Computer Science, Ashoka University
2025 **Teaching Assistantship Excellence Award**, Department of Computer Science, Ashoka University
2024 **Summa Cum Laude**, Ashoka University
2024 **Silver Medalist**, Department of Computer Science, Ashoka University
2024 **Builder's Award for Service Excellence**, Department of Computer Science, Ashoka University
2022 **Undergraduate Research Excellence Award**, Department of Mathematics, Ashoka University

OTHER EXPERIENCES

Ashoka University <i>8 × Teaching Assistant</i> - TA for multiple courses like Information Security, Data Structures, Discrete Mathematics (feedback: 4.88/5) - Facilitated the Science Communication module at the Lodha Genius Program for 200+ high school students	Sonipat, India <i>Aug 2022 – May 2025</i>
Plaksha University <i>Instructor</i> - Taught a Microcontrollers course to 200+ high school students through IoT projects using ESP32 chipset - Designed a game development module, building 5 mathematical and hardware games for hands-on learning	Mohali, India <i>April 2023 – June 2023</i>
Lehigh University <i>Exchange Student</i> - Consulted for Global Good Fund, establishing metrics to evaluate ESG impact, measure non-monetary outcomes - Strategized optimal investment approaches to enhance ESG goals and ensure measurable financial outcomes	Bethlehem, US <i>May 2022 – Jun 2022</i>
WebVeda <i>Tech and Product Manager</i> - Ideated and developed the platform, scaling to 300k learners and generating \$1M revenue within 10 months - Managed tech infrastructure, including payment gateways, email integration, and other critical components	Remote <i>Feb 2021 – April 2022</i>

OTHER ACTIVITIES

Workshop on Lattice-based Post-quantum Cryptography <i>Speaker, Student Organiser</i>	Ashoka University <i>April 2024</i>
Winter and Summer Schools in Cryptography and Security <i>Selected Participant</i>	Multiple Locations <i>2023 – 2025</i>
Academic Affairs Board <i>Computer Science Student Representative</i>	Ashoka University <i>May 2023 – May 2024</i>
IEEE Ashoka Student Branch <i>Director of Technology</i>	Ashoka University <i>Aug 2023 – May 2024</i>
Computer Science Society <i>Student Advisor, Interim President</i>	Ashoka University <i>Aug 2022 – May 2024</i>
Program in Mathematics for Young Scientists <i>Mehta Fellow</i>	Boston, US <i>Jul 2020 – Aug 2021</i>

RELEVANT COURSEWORK

Trustworthy AI, Information and Coding Theory, Computer Security and Privacy, Lattice-based Cryptography, Foundations of Dilithium, Blockchain and Cryptocurrencies, Quantum Computing, Symbolic Logic, Theory of Computation, Games on Graphs

TECHNICAL SKILLS

Languages: TypeScript, Java, Python, C/C++, SQL, HTML, SCSS, Solidity, Assembly, SML
Technologies: Qiskit, Hyperledger Fabric, Git, L^AT_EX, Docker, SageMath, Markdown
Other: Arduino Uno, ESP32, Raspberry Pi Pico